which proves the statement presented in Equation (24) with $C_4'(d,s,R) \coloneqq T_s^2 R^2 d^{1+s}$.

We now consider the constant $C_3(M,d)$ from Theorem 5. Suppose we have

$$
\begin{aligned}
\|\psi\|_2^2 &\geq \frac{T_s^2 R^2 d^{1+s}}{1-S^2}\sum_{i=1}^d \lambda_i^{2s} + \frac{C_3}{(1-S^2)}\frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n)\sqrt{\lambda_1\cdots\lambda_d}}\\
(1-S^2)\|\psi\|_2^2 &\geq T_s^2 R^2 d^{1+s}\sum_{i=1}^d \lambda_i^{2s} + C_3\frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n)\sqrt{\lambda_1\cdots\lambda_d}}\\
\|\psi\|_2^2 &\geq S^2\|\psi\|_2^2 + T_s^2 R^2 d^{1+s}\sum_{i=1}^d \lambda_i^{2s} + C_3\frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n)\sqrt{\lambda_1\cdots\lambda_d}}\\
\|\psi\|_2^2 &\geq \|\psi-\psi*\varphi_\lambda\|_2^2 + C_3\frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n)\sqrt{\lambda_1\cdots\lambda_d}}
\end{aligned}
$$

then, by Theorem 5, we can ensure that

$$
\mathbb{P}_{p\times q\times r}\left(\Delta_\alpha^{\lambda,B}(\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_B)=0\right)\leq\beta.
$$

By definition of uniform separation rates, we deduce that

$$
\begin{aligned}
\rho\left(\Delta_\alpha^{\lambda,B},\mathcal{S}_d^s(R),\beta,M\right)^2 &\leq \frac{T_s^2 R^2 d^{1+s}}{1-S^2}\sum_{i=1}^d \lambda_i^{2s} + \frac{C_3}{(1-S^2)}\frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n)\sqrt{\lambda_1\cdots\lambda_d}}\\
&\leq C_4(M,d,s,R,\beta)\left(\sum_{i=1}^d \lambda_i^{2s} + \frac{\ln\left(\frac{1}{\alpha}\right)}{(m+n)\sqrt{\lambda_1\cdots\lambda_d}}\right)
\end{aligned}
$$

for $C_4(M,d,s,R,\beta) \coloneqq \max\left\{\frac{T_s^2R^2d^{1+s}}{1-S^2},\frac{C_3(M,d)}{\beta(1-S^2)}\right\}$.

E.7 Proof of Corollary 7

By Theorem 6, if $\lambda_1\cdots\lambda_d\leq 1$, we have

$$
\rho\left(\Delta_\alpha^{\lambda,B},\mathcal{S}_d^s(R),\beta,M\right)^2 \leq C_4(M,d,s,R,\beta)\left(\sum_{i=1}^d \lambda_i^{2s} + \frac{\ln\left(\frac{1}{\alpha}\right)}{(m+n)\sqrt{\lambda_1\cdots\lambda_d}}\right).
$$

We want to express the bandwidths in terms of the sum of sample sizes $m+n$ raised to some negative power such that the terms $\sum_{i=1}^d \lambda_i^{2s}$ and $\frac{1}{(m+n)\sqrt{\lambda_1\cdots\lambda_d}}$ have the same behaviour in $m+n$. With the choice of bandwidths $\lambda_i^* \coloneqq (m+n)^{-2/(4s+d)}$ for $i=1,\ldots,d$, the term $\sum_{i=1}^d(\lambda_i^*)^{2s}$ has order $(m+n)^{-4s/(4s+d)}$ and the term $\frac{1}{(m+n)\sqrt{\lambda_1^*\cdots\lambda_d^*}}$ has order $(m+n)^{d/(4s+d)-1} = (m+n)^{-4s/(4s+d)}$. So, indeed, this choice of bandwidths leads to the same behaviour in $m+n$ for the two terms, which gives the smallest order of $m+n$ possible.

It is clear that $\lambda_1^* \cdots \lambda_d^* < 1$, we find that

$$\begin{aligned}
\rho\Big(\Delta_\alpha^{\lambda^*,B}, \mathcal{S}_d^s(R), \beta, M\Big)^2 &\leq C_4(M,d,s,R,\beta)\left(\sum_{i=1}^d (\lambda_i^*)^{2s} + \frac{\ln\big(\frac{1}{\alpha}\big)}{(m+n)\sqrt{\lambda_1^*\cdots\lambda_d^*}}\right) \\
&\leq C_4(M,d,s,R,\beta)\left((m+n)^{-4s/(4s+d)} + \ln\left(\frac{1}{\alpha}\right)(m+n)^{-4s/(4s+d)}\right) \\
&\leq C_4(M,d,s,R,\beta)\ln\left(\frac{1}{\alpha}\right)(m+n)^{-4s/(4s+d)} \\
&= C_5(M,d,s,R,\alpha,\beta)^2\,(m+n)^{-4s/(4s+d)}
\end{aligned}$$

for $C_5(M,d,s,R,\alpha,\beta) \coloneqq \sqrt{C_4(M,d,s,R,\beta)\ln\big(\frac{1}{\alpha}\big)}$. We deduce that

$$\rho\Big(\Delta_\alpha^{\lambda^*,B}, \mathcal{S}_d^s(R), \beta, M\Big) \leq C_5(M,d,s,R,\alpha,\beta)\,(m+n)^{-2s/(4s+d)}.$$

E.8 Proof of Proposition 8

By definition of $u_\alpha^{B_2}$, we have

$$\frac{1}{B_2}\sum_{b=1}^{B_2} \mathbb{1}\left(\max_{\lambda\in\Lambda}\left(\widehat{M}_{\lambda,2}^{\,b}\Big(\mu^{(b,2)}\big|\mathbb{X}_m,\mathbb{Y}_n\Big) - \widehat{q}^{\,\lambda,B_1}_{1-u_\alpha^{B_2}\left(\mathbb{Z}_{B_2}\middle|\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1}\right)w_\lambda}\Big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m,\mathbb{Y}_n\Big)\right) > 0\right) \leq \alpha.$$

Taking the expectation on both sides, we get

$$\mathbb{P}_{p\times p\times r\times r}\left(\max_{\lambda\in\Lambda}\left(\widehat{\mathrm{MMD}}^2_\lambda(\mathbb{X}_m,\mathbb{Y}_n) - \widehat{q}^{\,\lambda,B_1}_{1-u_\alpha^{B_2}\left(\mathbb{Z}_{B_2}\middle|\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1}\right)w_\lambda}\Big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m,\mathbb{Y}_n\Big)\right) > 0\right) \leq \alpha$$

as under the null hypothesis $\mathcal{H}_0\colon p = q$, we have $\Big(\widehat{M}_{\lambda,2}^{\,b}\big(\mu^{(b,2)}\big|\mathbb{X}_m,\mathbb{Y}_n\big)\Big)_{1\leq b\leq B_2}$ distributed like $\widehat{\mathrm{MMD}}^2_\lambda(\mathbb{X}_m,\mathbb{Y}_n)$. Using the bisection search approximation which satisfies

$$\widehat{u}_\alpha^{B_{2:3}}\big(\mathbb{Z}_{B_2}\big|\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1}\big) \leq u_\alpha^{B_2}\big(\mathbb{Z}_{B_2}\big|\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1}\big),$$

we get

$$\begin{aligned}
&\mathbb{P}_{p\times p\times r\times r}\left(\max_{\lambda\in\Lambda}\left(\widehat{\mathrm{MMD}}^2_\lambda(\mathbb{X}_m,\mathbb{Y}_n) - \widehat{q}^{\,\lambda,B_1}_{1-\widehat{u}_\alpha^{B_{2:3}}\left(\mathbb{Z}_{B_2}\middle|\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1}\right)w_\lambda}\Big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m,\mathbb{Y}_n\Big)\right) > 0\right) \\
\leq\ &\mathbb{P}_{p\times p\times r\times r}\left(\max_{\lambda\in\Lambda}\left(\widehat{\mathrm{MMD}}^2_\lambda(\mathbb{X}_m,\mathbb{Y}_n) - \widehat{q}^{\,\lambda,B_1}_{1-u_\alpha^{B_2}\left(\mathbb{Z}_{B_2}\middle|\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1}\right)w_\lambda}\Big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m,\mathbb{Y}_n\Big)\right) > 0\right) \\
\leq\ &\alpha.
\end{aligned}$$

We deduce that

$$\mathbb{P}_{p\times p\times r\times r}\Big(\Delta_\alpha^{\Lambda^w,B_{1:3}}(\mathbb{X}_m,\mathbb{Y}_n,\mathbb{Z}_{B_1},\mathbb{Z}_{B_2}) = 1\Big) \leq \alpha.$$

E.9 Proof of Theorem 9

Consider some $u^* \in (0,1)$ to be determined later. For $b = 1, \dots, B_2$, let

$$W_b\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\Big) := \mathbb{1}\left(\max_{\lambda\in\Lambda}\left(\widehat{M}^{\,b}_{\lambda,2}\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n\Big) - \widehat{q}^{\,\lambda,B_1}_{1-u^*w_\lambda}\big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m, \mathbb{Y}_n\big)\right) > 0\right),$$

so that, following a similar argument to the one presented in Appendix E.8, we obtain that $\mathbb{E}_{p\times q\times r\times r}\big[W_b\big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\big)\big]$ is equal to

$$\mathbb{P}_{p\times p\times r}\left(\max_{\lambda\in\Lambda}\left(\widehat{\mathrm{MMD}}^2_\lambda(\mathbb{X}_m, \mathbb{Y}_n) - \widehat{q}^{\,\lambda,B_1}_{1-u^*w_\lambda}\big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m, \mathbb{Y}_n\big)\right) > 0\right).$$

Consider the events

$$\mathcal{A}' := \left\{\frac{1}{B_2}\sum_{b=1}^{B_2} W_b\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\Big) - \mathbb{E}_r\Big[W_b\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\Big)\Big] \le \sqrt{\frac{1}{2B_2}\ln\left(\frac{2}{\beta}\right)}\right\}$$

and

$$\begin{aligned}\mathcal{A} := \Bigg\{\frac{1}{B_2}\sum_{b=1}^{B_2} W_b\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\Big) &- \mathbb{E}_{p\times q\times r\times r}\Big[W_b\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\Big)\Big] \\ &\le \sqrt{\frac{1}{2B_2}\ln\left(\frac{2}{\beta}\right)}\Bigg\}.\end{aligned}$$

Using Hoeffding's inequality, we obtain that $\mathbb{P}_r\big(\mathcal{A}'\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\big) \ge 1 - \frac{\beta}{2}$ for any $\mathbb{X}_m$, $\mathbb{Y}_n$ and $\mathbb{Z}_{B_1}$, we deduce that $\mathbb{P}_{p\times q\times r\times r}(\mathcal{A}) \ge 1 - \frac{\beta}{2}$.

First, assuming that the event $\mathcal{A}$ holds, we show that $u^{B_2}_\alpha\big(\mathbb{Z}_{B_2}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\big) \ge \alpha$. Since we assume that the event $\mathcal{A}$ holds, the bounds we obtain hold with probability $1 - \frac{\beta}{2}$. We have

$$\begin{aligned}&\frac{1}{B_2}\sum_{b=1}^{B_2}\mathbb{1}\left(\max_{\lambda\in\Lambda}\left(\widehat{M}^{\,b}_{\lambda,2}\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n\Big) - \widehat{q}^{\,\lambda,B_1}_{1-u^*w_\lambda}\big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m, \mathbb{Y}_n\big)\right) > 0\right)\\
&= \frac{1}{B_2}\sum_{b=1}^{B_2} W_b\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\Big)\\
&\le \mathbb{E}_{p\times p\times r\times r}\Big[W_b\Big(\mu^{(b,2)}\big|\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}\Big)\Big] + \sqrt{\frac{1}{2B_2}\ln\left(\frac{2}{\beta}\right)}\\
&= \mathbb{P}_{p\times p\times r}\left(\max_{\lambda\in\Lambda}\left(\widehat{\mathrm{MMD}}^2_\lambda(\mathbb{X}_m, \mathbb{Y}_n) - \widehat{q}^{\,\lambda,B_1}_{1-u^*w_\lambda}\big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m, \mathbb{Y}_n\big)\right) > 0\right) + \sqrt{\frac{1}{2B_2}\ln\left(\frac{2}{\beta}\right)}\\
&= \mathbb{P}_{p\times p\times r}\left(\bigcup_{\lambda\in\Lambda}\left\{\widehat{\mathrm{MMD}}^2_\lambda(\mathbb{X}_m, \mathbb{Y}_n) > \widehat{q}^{\,\lambda,B_1}_{1-u^*w_\lambda}\big(\mathbb{Z}_{B_1}\big|\mathbb{X}_m, \mathbb{Y}_n\big)\right\}\right) + \sqrt{\frac{1}{2B_2}\ln\left(\frac{2}{\beta}\right)}.\end{aligned}$$